

DMS201: Introduction to Management

Module-3: Operations Management

(INTRODUCTION TO MANAGEMENT)

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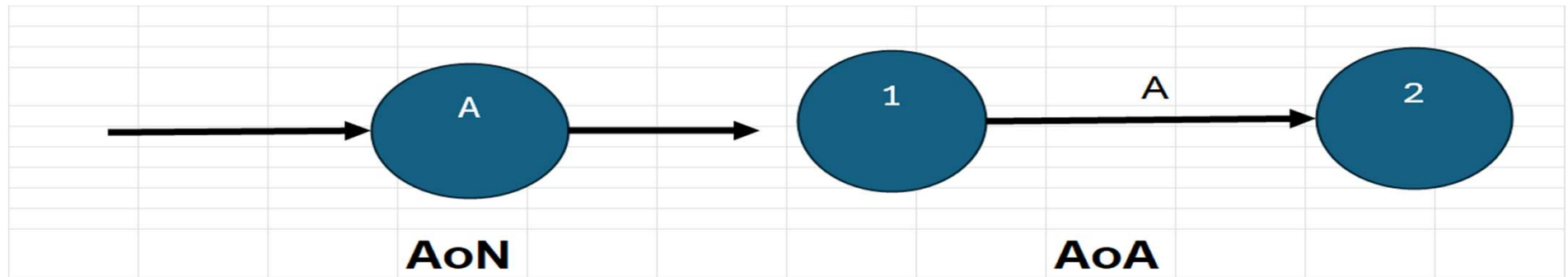
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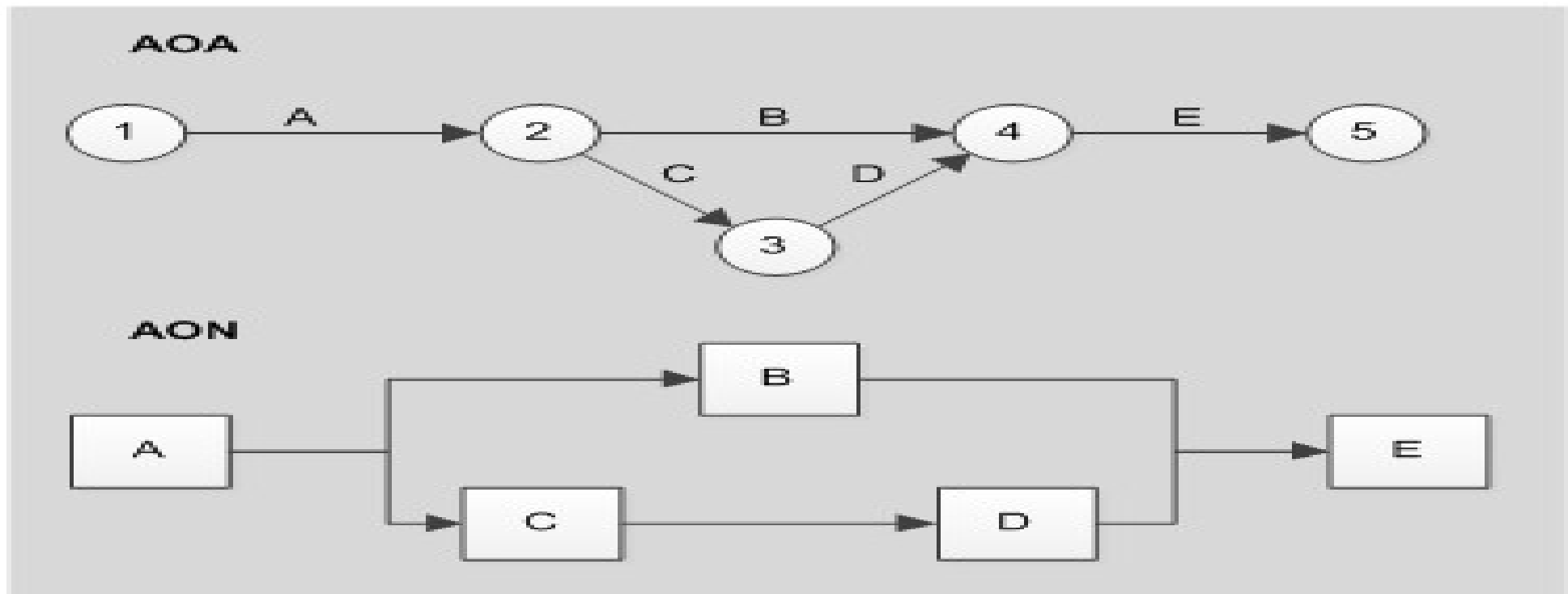


Analysis of Performance (AoA and AoN)

- Depending on the nomenclature we have two different ways of depicting an/a activity/job/task, i.e., either using a node or an arc.
- Hence we have **Activity on Arc (AOA)** and **Activity on Node (AON)** depiction of projects.



Analysis of Performance (AoA and AoN)

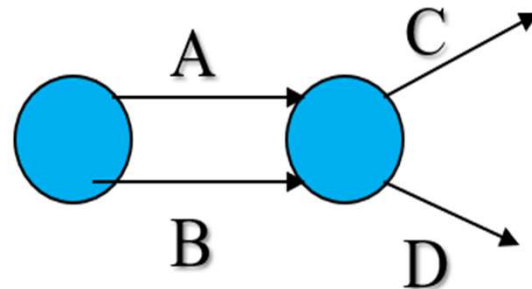


Analysis of Performance (AoA and AoN)

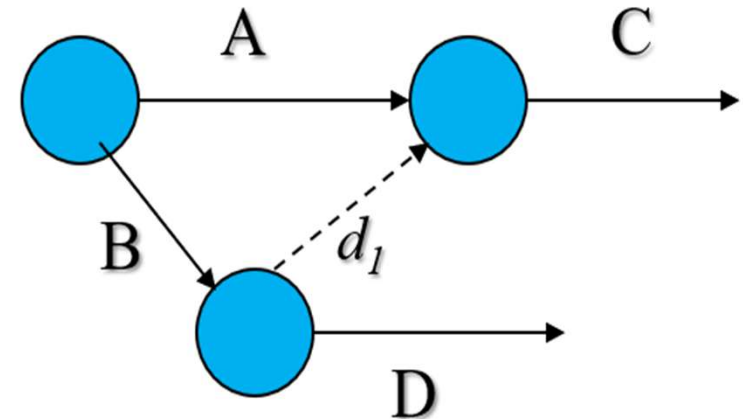
- Work **begins** on a single node and **ends** on a single node, for AoA.
- Two nodes are connected by **one and only one** activity (*edge* or *arc*).
- Thus,

<u>Act.</u>	<u>I.P.</u>
A	—
B	—
C	A,B
D	B

No, and why not?



Rather, this is preferred and why?



d_1 is **Dummy Activity**

Analysis of Performance

(Network: Important Assumptions/Rules)

- 1) All activities must be logically linked to each other; those that precede and other, subsequent activities.
- 2) Network diagrams usually flow from left to right.
- 3) An activity cannot begin until all preceding connected activities have been completed.
- 4) Arrows on networks indicate precedence and logical flow. Arrows can cross over each other, although it is helpful for clarity's sake to limit this effect when possible.

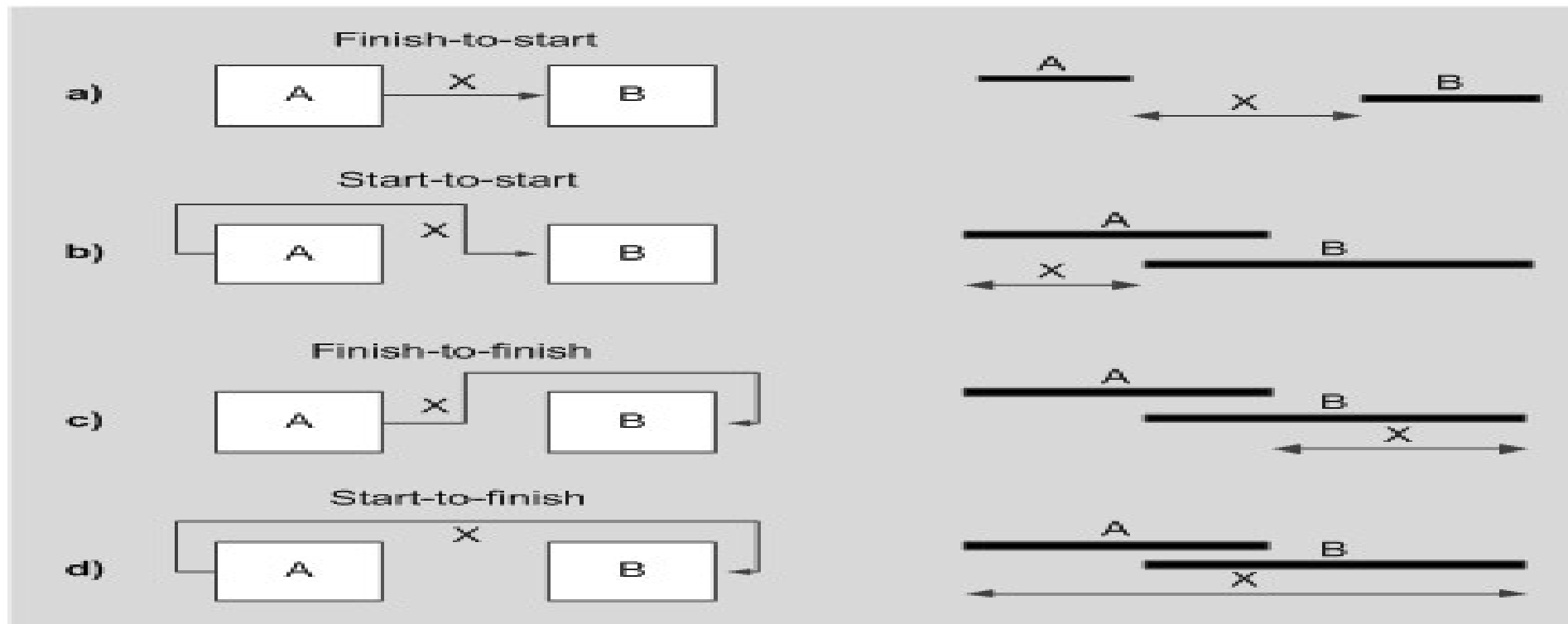
Analysis of Performance

(Network: Important Assumptions/Rules)

- 5) Each activity should have a unique identifier associated with it (number, letter, code, etc.). For simplicity, these identifiers should occur in ascending order; each one should be larger than the identifiers of preceding activities.
- 6) Looping, or recycling through activities, is not permitted.
- 7) Although not required, it is common to start a project from a single beginning node, even in the case when multiple start points are possible. A single node point also is typically used as a project end indicator.

Analysis of Performance

(Network: Relationship between Jobs)



Analysis of Performance

(Network: Relationship between Jobs)

- **Finish-to-start** relationship (FS): The Activity B may not start until x days after activity A has finished. By letting $x = 0$, the precedence relationship becomes “B may start as soon as A is finished”.
- **Start-to-start** relationship (SS). The Activity B may not start until x days after the start of activity A.
- **Finish-to-finish** relationship (FF). The Activity B may not finish earlier than x days after activity A is finished.
- **Start-to-finish** relationship (SF). The Activity B may not finish earlier than x days after activity A has started.

Analysis of Performance (**Network: Relationship between Jobs**)

- In order to perform a **forward scheduling** we have to “normalize” the FS, FF and SF time relationships to a SS relationship.
- Correspondingly, to perform a **backward scheduling** we “normalize” the time relationships FS, SS and SF to a FF relationship.

Analysis of Performance

(Network: Relationship between Jobs)

Type	Forward	Backward
FS	$d_A + x$	$d_B + x$
SS	x	$d_B + x - d_A$
FF	$d_A + x - d_B$	x
SF	$x - d_B$	$x - d_A$

Analysis of Performance

(Network: Important Rules)

- The time relationships (a time interval for example number of days) can both be a *lower bound* or an *upper bound* on the time that separates the start/finish of two next following activities.
- Furthermore, time points and time relationships can both be positive or negative depending on we calculate forward or backward in time or reverse the time axis.
- There may be two approaches to assign time interval to networks:
 - ❖ Time interval is a *lower or upper bound* on *event* times of activities, i.e. a time relationships
 - ❖ Time interval is the *duration* of an *activity*, i.e. an attribute.

Analysis of Performance (Network: Calculations)

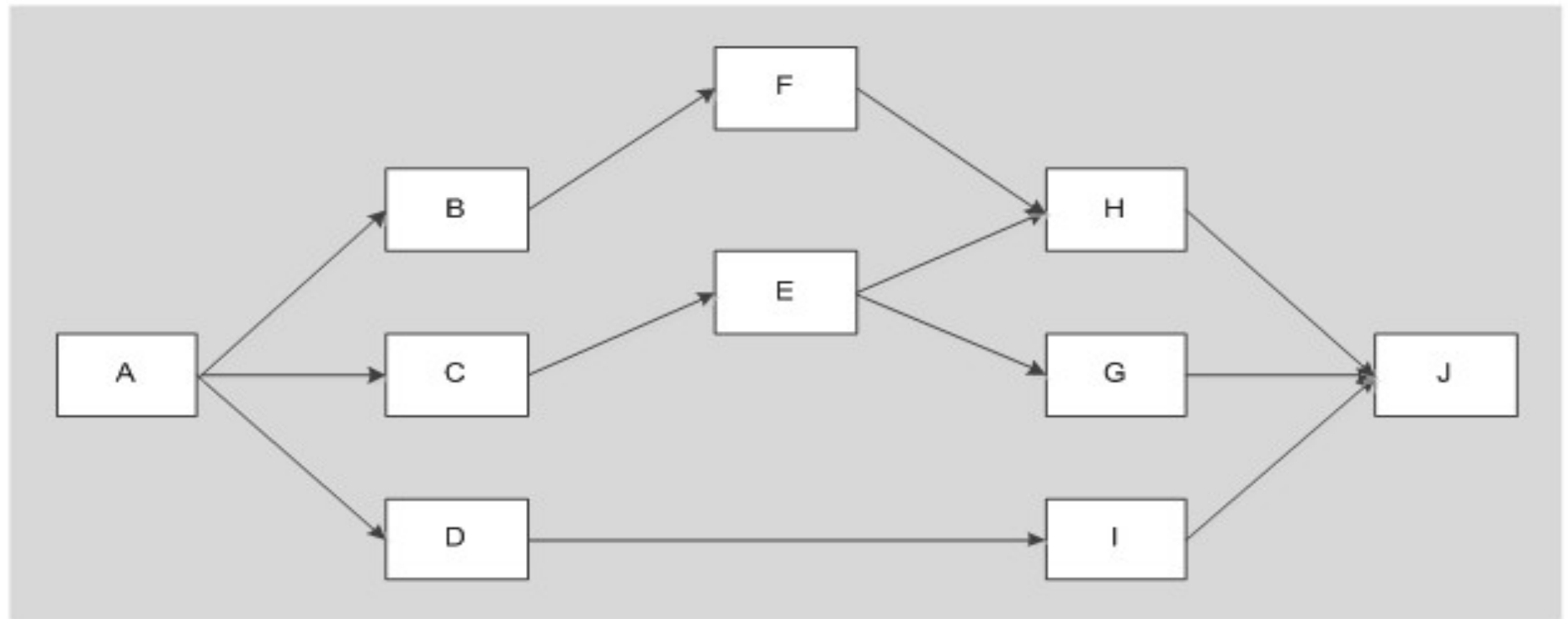
- Early Start (ES_j) = $\max_{\forall i}(EF_i)$, $i < j$
- Early Finish (EF_j) = $ES_j + d_j$
- Late Finish (LF_j) = $\min_{\forall i}(LS_i)$
- Late Start (LS_j) = $LF_j - d_j$, $i < j$
- Total Float (TF_i) = $LS_i - ES_i = LF_i - LS_i$
- Free Float (FF_i) = $\min_{\forall j}(ES_j) - EF_i$
- Total Float (TF): Amount of time a task can be delayed without delaying the project finish date and for the **Critical Path (CP) it is zero**
- Free Float (FF): Amount of time a task can be delayed without delaying the start of the next task.
- Free Float (FF) can never be greater than Total Float (TF), i.e., $FF \leq TF$

Analysis of Performance

(CPM: Example # 28)

Jobs	Precedence	Days (d_i)	t_{i-j}
A	None	4	0
B	A	4	0
C	A	2	0
D	A	5	0
E	C	2	0
F	B	4	0
G	E	2	0
H	E, F	2	0
I	D	3	0
J	G, H, I	4	0

Analysis of Performance (CPM: Example # 28)



Analysis of Performance

(CPM: Example # 28)

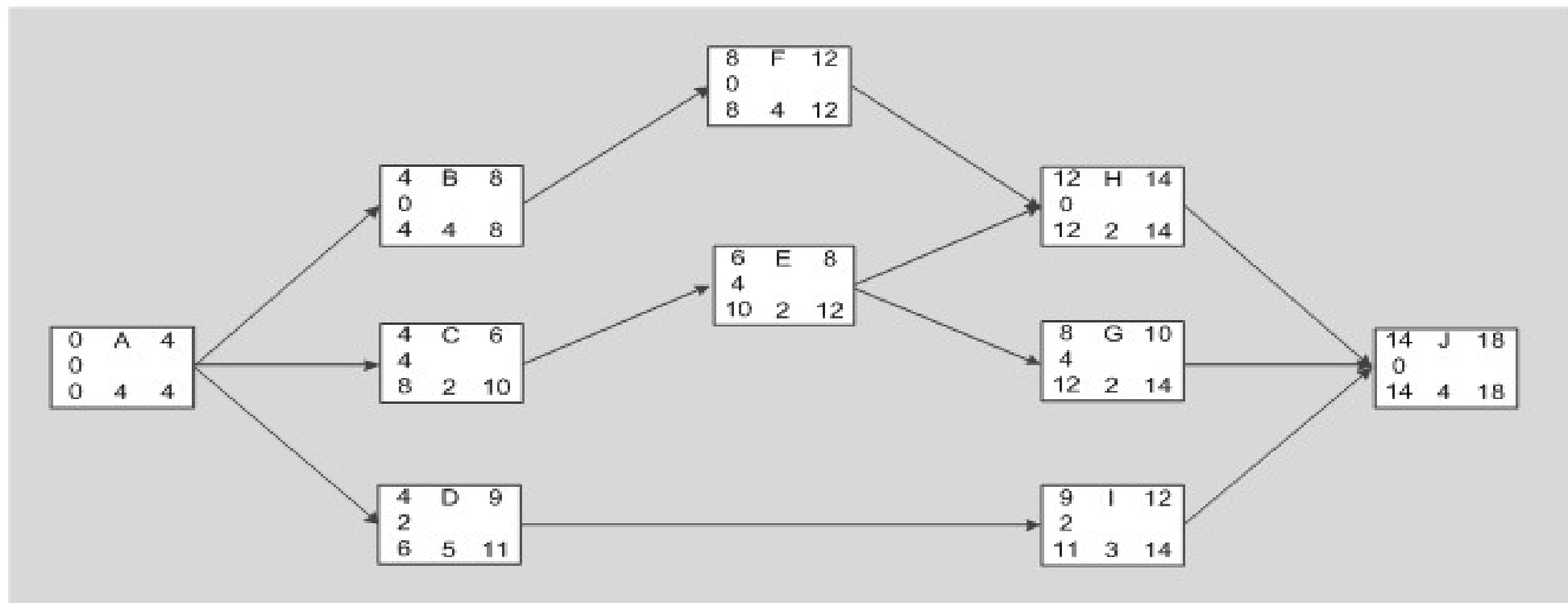
Jobs	Precedence	Days (d_i)	t_{i-j}	ES	EF	LF	LS
A	None	4	0	0	$0+4=4$	$\text{Min}(4, 8, 6) = 4$	$4-4=0$
B	A	4	0	4	$4+4=8$	8	$8-4=4$
C	A	2	0	4	$4+2=6$	10	$10-2=8$
D	A	5	0	4	$4+5=9$	11	$11-5=6$
E	C	2	0	6	$6+2=8$	12	$12-2=10$
F	B	4	0	8	$8+4=12$	12	$12-4=8$
G	E	2	0	8	$8+2=10$	14	$14-2=12$
H	E, F	2	0	$\text{Max}(8,12)=12$	$12+2=14$	14	$14-2=12$
I	D	3	0	9	$9+3=12$	14	$14-3=11$
J	G, H, I	4	0	$\text{Max}(14,10,12)=14$	$14+4=18$	18	$18-4=14$

Analysis of Performance

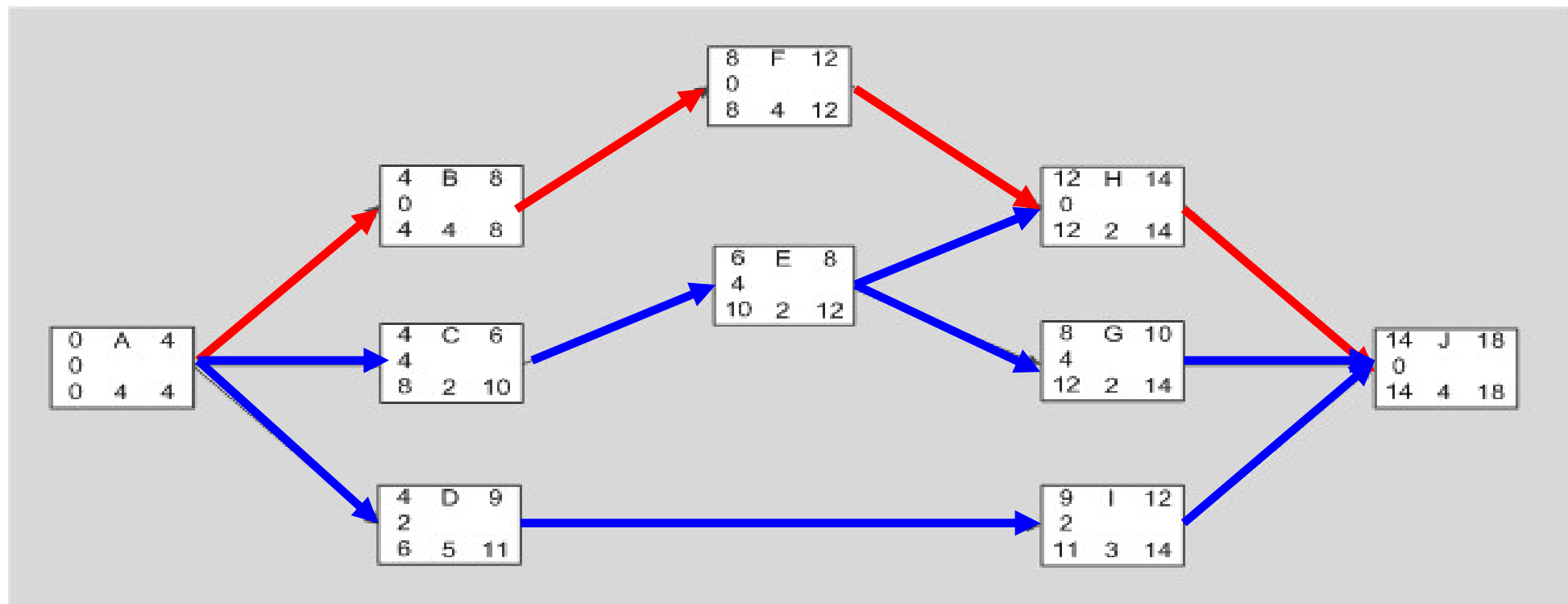
(CPM: Example # 28)

Jobs	ES_i	EF_i	LF_i	LS_i	$TF_i = LS_i - ES_i = LF_i - EF_i$	$FF_i = \min_{\forall j} (ES_j) - EF_i \leq TF_i$	CPM iff $TF_i = 0$
A	0	$0+4=4$	$\text{Min}(4, 8, 6) = 4$	$4-4=0$	$0-0=4-4=0$	$4-4=0$	Yes
B	4	$4+4=8$	8	$8-4=4$	$4-4=8-8=0$	$8-8=0$	Yes
C	4	$4+2=6$	10	$10-2=8$	$8-4=10-6=4$	$6-6=0$	No
D	4	$4+5=9$	11	$11-5=6$	$6-4=11-9=2$	$9-9=0$	No
E	6	$6+2=8$	12	$12-2=10$	$10-6=12-8=4$	$\text{Min}(8,12)-8=8-8=0$	No
F	8	$8+4=12$	12	$12-4=8$	$8-8=12-12=0$	$12-12=0$	Yes
G	8	$8+2=10$	14	$14-2=12$	$12-8=14-10=4$	$14-10=4$	No
H	$\text{Max}(8,12)=12$	$12+2=14$	14	$14-2=12$	$12-12=14-14=0$	$14-14=0$	Yes
I	9	$9+3=12$	14	$14-3=11$	$11-9=14-12=2$	$14-12=2$	No
J	$\text{Max}(14,10,12)=14$	$14+4=18$	18	$18-4=14$	$14-14=18-18=0$	$18-18=0$	Yes

Analysis of Performance (CPM: Example # 28)



Analysis of Performance (CPM: Example # 28)



Analysis of Performance (CPM: Example # 28)

- **A – B – F – H – J = 18**
- A – C – E – H – J = 14
- A – C – E – G – J = 14
- A – D – I – J = 16

Analysis of Performance (PERT)

- PERT was originally conceived in the late 1950's as a joint venture of the U.S. Navy and Booz Allen Hamilton for the Polaris missile program.
- This technique uses probabilistic estimation; that is, it assigns likely probabilities to a range of estimates for each activity that reflect the uncertainties inherent in the estimation process.

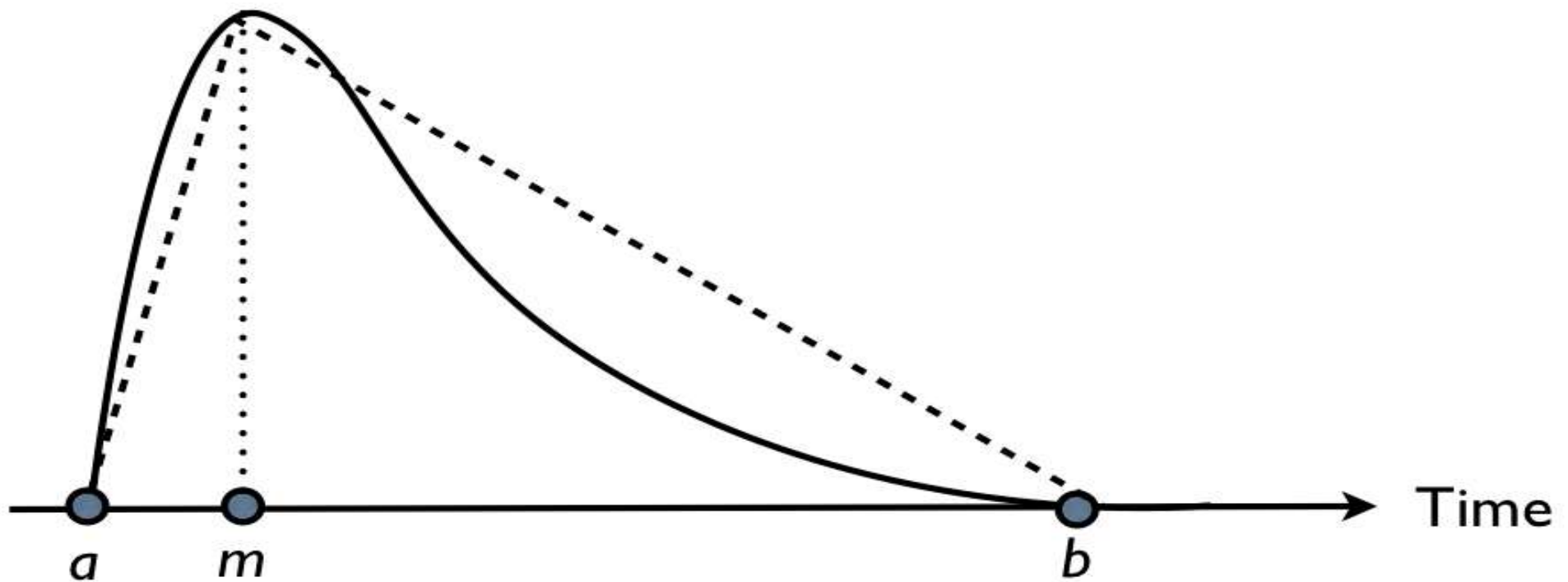
Analysis of Performance (PERT)

- PERT estimates recognize that for any activity duration, there are three important estimates:
- **Optimistic time** (a) – the time an activity will take if everything happens as planned. In estimating the duration for a , there should be only a 1% chance (1/100) that the activity time will be $<a$.
- **Most likely time** (m) – the most realistic time to complete the activity.
- **Pessimistic time** (b) – the time an activity will take if nothing happens as planned; that is, if everything goes wrong or conditions are most unfavorable. In estimating the duration for b , there should be only a 1% chance (1/100) that the activity time will be $>b$.

Analysis of Performance (PERT)

- The logic for this assumption is based on the understanding that to achieve a probability distribution with a 99% confidence interval, observations should lie within three standard deviations of the mean in either direction.
- A spread of six standard deviations from tail to tail in the probability distribution, then, accounts for 99.7% of the possible activity duration alternatives.

Analysis of Performance (PERT)



Analysis of Performance (PERT)

- The beta distribution suggests the formulae for
- $E(T) = \text{Expected Time} = (a + 4 \times m + b)/6$
- $\text{Var}(T) = \text{Variance of Time} = (b - a)/6\}^2$

Where

- ❖ a = Most optimistic time to complete the activity
- ❖ m = Most likely time to complete the activity, the mode of the distribution
- ❖ b = Most pessimistic time to complete the activity

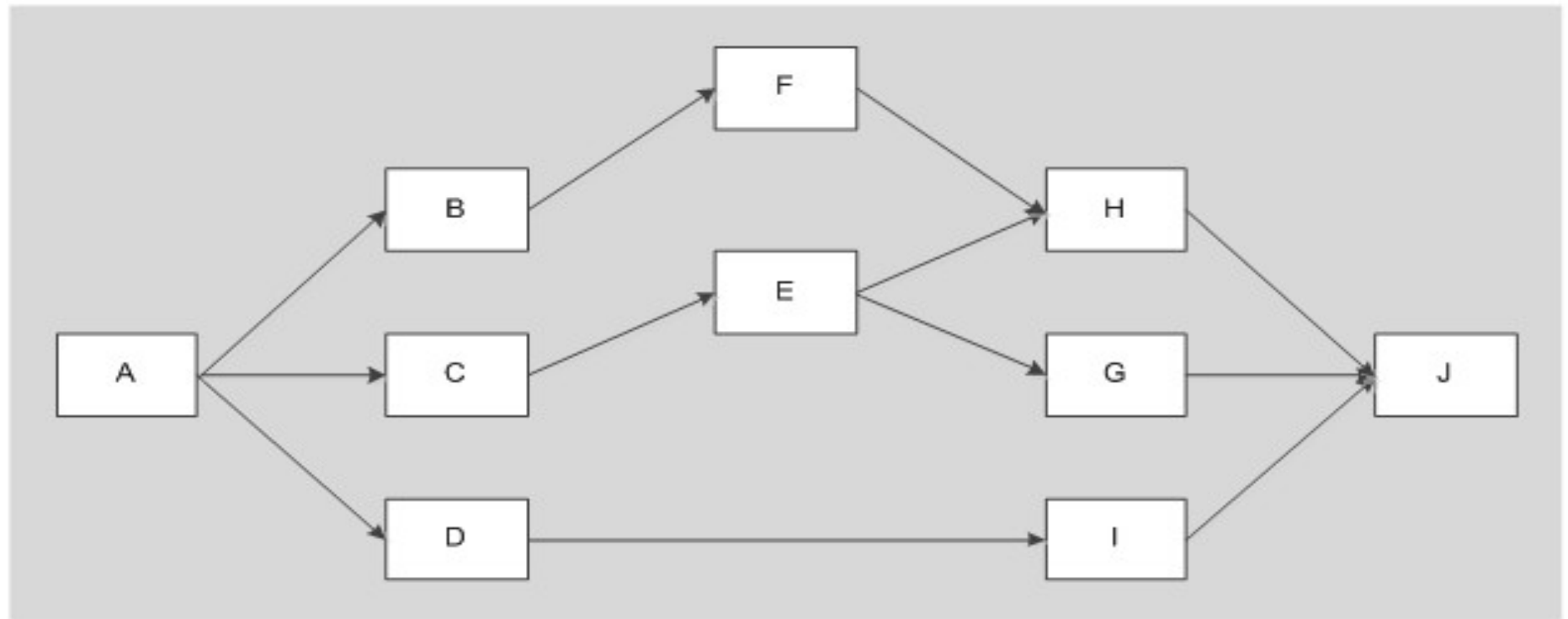
Analysis of Performance (PERT)

- The expected number of days required to finish a project is the sum of the number of days required for each activity/job/task.
- Remember here the time required for each **activity/job/task** must be its respective $E(t_i)$, where $i=1, \dots, m$ (the number of the activity/job/task)
- The critical path is also calculated using the respective expected time for each activity/job/task, i.e., $E(CP) = \{E(t_1) + E(t_2) + \dots + E(t_K)\}$, where 1, 2, ..., K are the **activities/jobs/tasks** in the **critical path (CP)**.

Analysis of Performance (PERT)

- Remember for finding out the variance of the total project we simply add the variances of the activities/jobs/tasks and not the standard deviation.
- The important assumption which helps us to add the variances, is that the activities/jobs/tasks are **independent**.
- Thus $\text{Var}(\text{CP}) = \{\text{Var}(t_1) + \text{Var}(t_2) + \dots + \text{Var}(t_k)\}$, where 1, 2, ..., K are the **activities/jobs/tasks** in the **critical path (CP)**.

Analysis of Performance (PERT: Example # 29)



Analysis of Performance

(PERT: Example # 29)

Jobs	Precedence	t_a	t_m	t_b	$E(t_i)$	$V(t_i)$	ES	EF	LF	LS
A	None	2	5	10	5.33	1.78	0	5.33	9.50	4.17
B	A	3	7	11	7.00	1.78	5.33	12.33	16.50	9.50
C	A	4	8	15	8.50	3.36	5.33	13.83	18.33	9.83
D	A	2	7	17	7.83	6.25	5.33	13.17	19.33	11.50
E	C	1	2	4	2.17	0.25	13.83	16.00	20.50	18.33
F	B	1	4	7	4.00	1.00	12.33	16.33	20.50	16.50
G	E	5	8	14	8.50	2.25	16.00	24.50	24.50	16.00
H	E, F	2	4	6	4.00	0.44	16.33	20.33	24.50	20.50
I	D	2	5	9	5.17	1.36	13.17	18.33	24.50	19.33
J	G, H, I	4	7	20	8.67	7.11	24.5	33.17	33.17	24.50

Analysis of Performance

(PERT: Example # 29)

Path (P_i)	$E(P_i)$	$Var(P_i)$	$D_1=30$ (% Complete)	$D_2=34$ (% incomplete)
A – B – F – H – J	29.00	12.11	$Pr(P_i \leq 30)$ $Pr\{(P_i - 29.00)/\sqrt{12.11} \leq (30 - 29.00)/\sqrt{12.11}\}$ $Pr(z_i \leq +0.28736) = 61.31\%$	$Pr(P_i \geq 34)$ $Pr\{(P_i - 29.00)/\sqrt{12.11} \geq (34 - 29.00)/\sqrt{12.11}\}$ $Pr(z_i \geq +1.43681) = 7.54\%$
A – C – E – H – J	28.67	12.94	$Pr(P_i \leq 30)$ $Pr\{(P_i - 28.67)/\sqrt{12.94} \leq (30 - 28.67)/\sqrt{12.94}\}$ $Pr(z_i \leq +0.36973) = 64.42\%$	$Pr(P_i \geq 34)$ $Pr\{(P_i - 28.67)/\sqrt{12.94} \geq (34 - 28.67)/\sqrt{12.94}\}$ $Pr(z_i \geq +1.48170) = 6.92\%$
A – C – E – G – J	33.17	14.75	$Pr(P_i \leq 30)$ $Pr\{(P_i - 33.17)/\sqrt{14.75} \leq (30 - 33.17)/\sqrt{14.75}\}$ $Pr(z_i \leq -0.82540) = 20.46\%$	$Pr(P_i \geq 34)$ $Pr\{(P_i - 33.17)/\sqrt{14.75} \geq (34 - 33.17)/\sqrt{14.75}\}$ $Pr(z_i \geq +0.21611) = 41.45\%$
A – D – I – J	27.00	16.50	$Pr(P_i \leq 30)$ $Pr\{(P_i - 27.00)/\sqrt{16.50} \leq (30 - 27.00)/\sqrt{16.50}\}$ $Pr(z_i \leq +0.73885) = 76.99\%$	$Pr(P_i \geq 34)$ $Pr\{(P_i - 27.00)/\sqrt{16.50} \geq (34 - 27.00)/\sqrt{16.50}\}$ $Pr(z_i \geq +1.72328) = 4.24\%$

Operations Management

- Introduction to production management
- Capacity planning and process analysis
- Network Analysis
- **Operations Research**

Operations Research

Operations Research

- The mathematical methods used to solve a resource allocation problem is called Operations Research
- In the mathematical modeling there are
 - ❖ **Objective Function(s):** $f_k(x), k=1, \dots, K$
 - ❖ **Decision Variables:** $x = x_1, \dots, x_n$
 - ❖ **Constraints:** $g_j(x) \geq b_j, j=1, \dots, m$

Operations Research

- A typical optimization problem in a single object function is of the form

$$\begin{aligned} &\text{Max/Min: } f(x) \\ &\text{s.t. } g_j(x) \geq b_j, j=1, \dots, m \\ &\quad x = x_1, \dots, x_n \end{aligned}$$

Operations Research

- **Static:** Decision variables do not involve sequences of decisions over multiple periods.
- **Dynamic:** Decision variables do involve sequences of decisions over multiple periods.
- **Deterministic:** Objective function and constraints.
- **Stochastic:** Objective function and/or constraints.

Operations Research

- $f(x)$ can be linear/non-linear, e.g., $2x_1 + 4x_2 - 15x_3$ or $\sin(x_1) + \log_e(x_2)$
- x can be continuous, integer, discrete, 0-1, etc.
- $g_j(x) \geq b_j, j=1, \dots, m$ can be linear/non-linear, e.g., $42x_1 + 9x_2 - 15x_3 \geq 103$ or $x_1^2 + x_2^{3/2} - 7x_3^{-4} \geq -\log_e(x_1^3 x_2/x_2)$

Operations Research

- Observe the system
- Formulate the mathematical model
- Solve and check the alternatives
- Check the validity of the results
- Implement and evaluate its implications

Operations Research (**Example # 28**)

- **Maximize:** $f(x) = x^2$, and $f_{\max}(x) = \infty$ with $x_{\max}$ being unbounded
- **Minimize:** $f(x) = x^2$, $-5 \leq x \leq -2$, and $f_{\min}(x) = +4$ with $x_{\min} = -2$

Operations Research (**Example # 28**)

- **Maximize:** $f(x) = 3x_1 + 5x_2$, $x_1 \in \mathbb{R}^+$, $x_2 \in \mathbb{R}^+$, and $f_{\max}(x) = \infty$ with $x_{1,\max}$, $x_{2,\max}$ both being unbounded
- **Minimize:** $f(x) = 2x_1 - 11x_2$, s.t.: $2x_1 + x_2 = 4$, $x_1 \in \mathbb{R}^+$, $x_2 \in \mathbb{R}^+$, and is $f_{\min}(x) = -44$ with $x_{1,\min} = 0$, $x_{2,\min} = +4$

Operations Research (**Example # 29**)

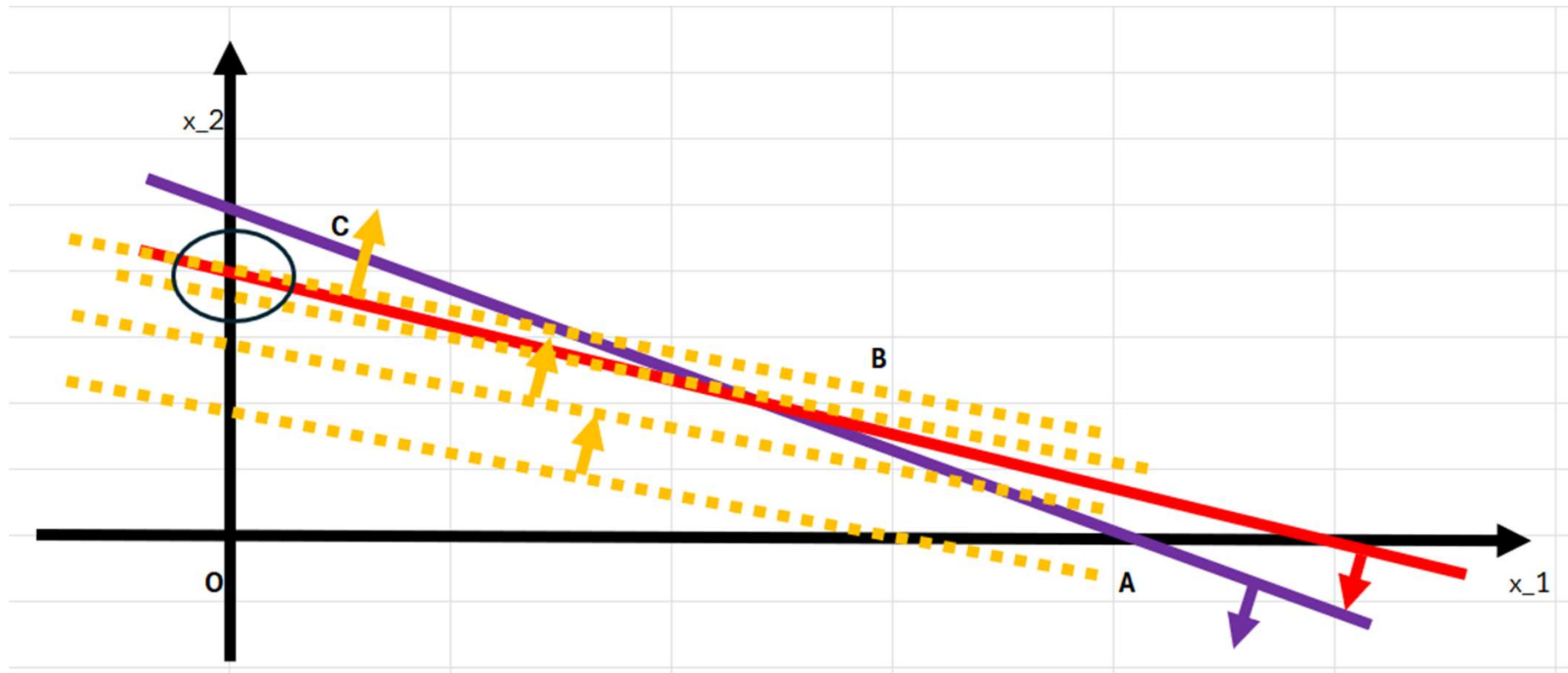
$$\text{Maximize: } f(x) = 2x_1 + 3x_2$$

$$\text{s.t.: } +4x_1 + 5x_2 \leq +20$$

$$+5x_1 + 4x_2 \leq +20$$

$$x_1 \in \mathbb{R}^+, x_2 \in \mathbb{R}^+$$

Operations Research (**Example # 29**)



Operations Research (**Example # 29**)

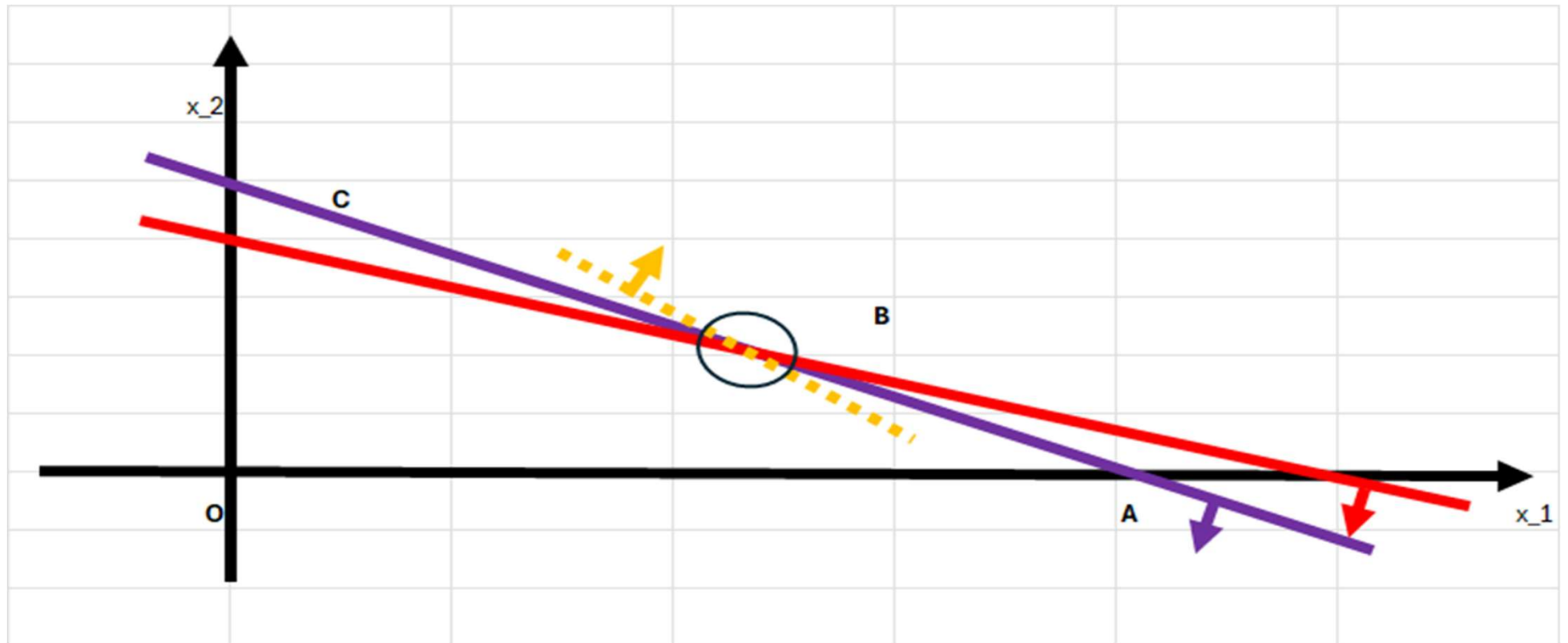
- O $(x_1=0, x_2=0), f(x_1=0, x_2=0)= 2 \times x_1 + 3 \times x_2 = 2 \times 0 + 3 \times 0 = 0$
- A $(x_1=4, x_2=0), f(x_1=4, x_2=0)= 2 \times x_1 + 3 \times x_2 = 2 \times 4 + 3 \times 0 = +8$
- B $(x_1=20/9, x_2=20/9), f(x_1=20/9, x_2=20/9)= 2 \times x_1 + 3 \times x_2 = 2 \times 20/9 + 3 \times 20/9 = +100/9 = +11.11$
- C $(x_1=0, x_2=4), f(x_1=0, x_2=4)= 2 \times x_1 + 3 \times x_2 = 2 \times 0 + 3 \times 4 = +12.00$

Operations Research (**Type of Solutions**)

- **Unique Solution**: Single optimal solution for objective function
- **Alternative/Multiple Solutions**: Alternative/Multiple optimal solutions for objective function
- **Unbounded Solution**: Infinite optimal solution for objective function
- **Infeasible Solution**: Infeasible optimal solution for objective function
- **Degenerate Solution**: Degenerate optimal solution for objective function as some constraints are redundant

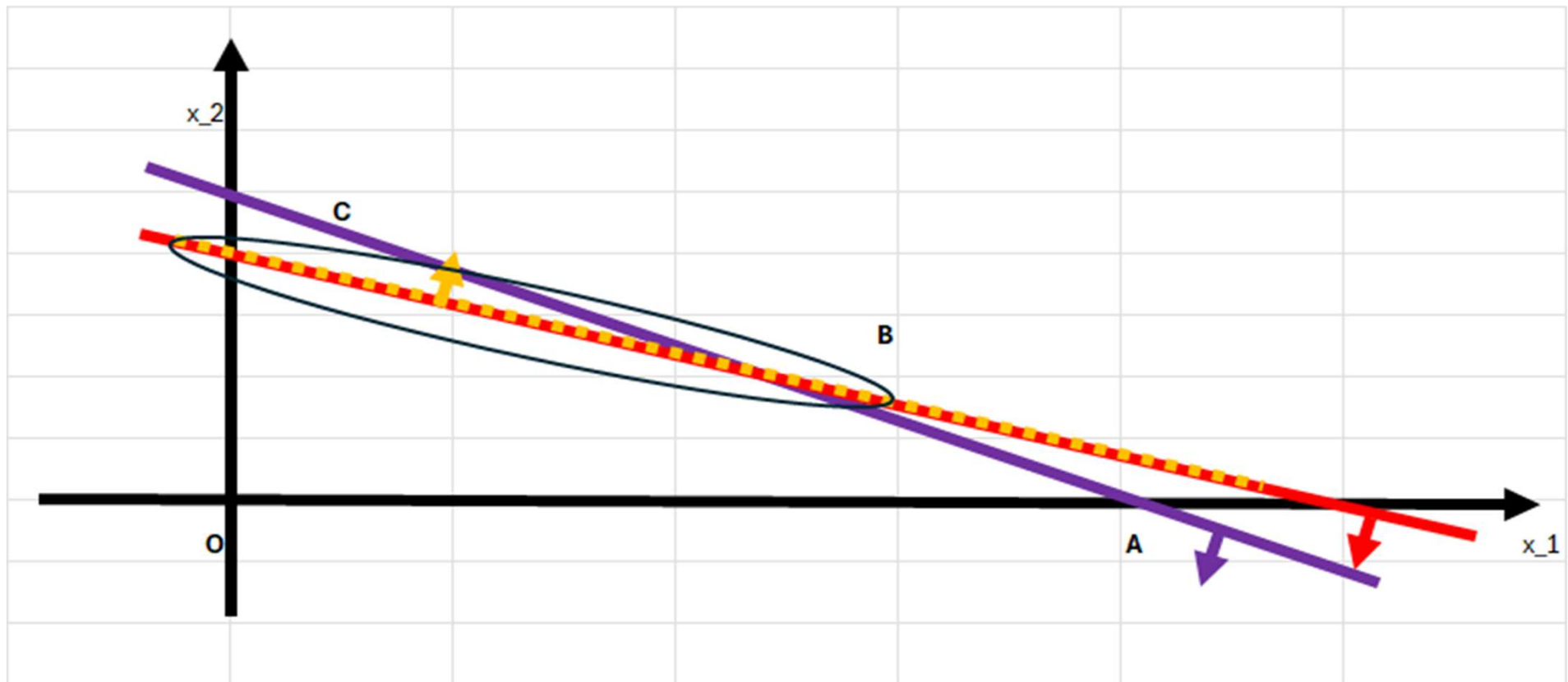
Operations Research

(Unique Solution: Maximization Case)



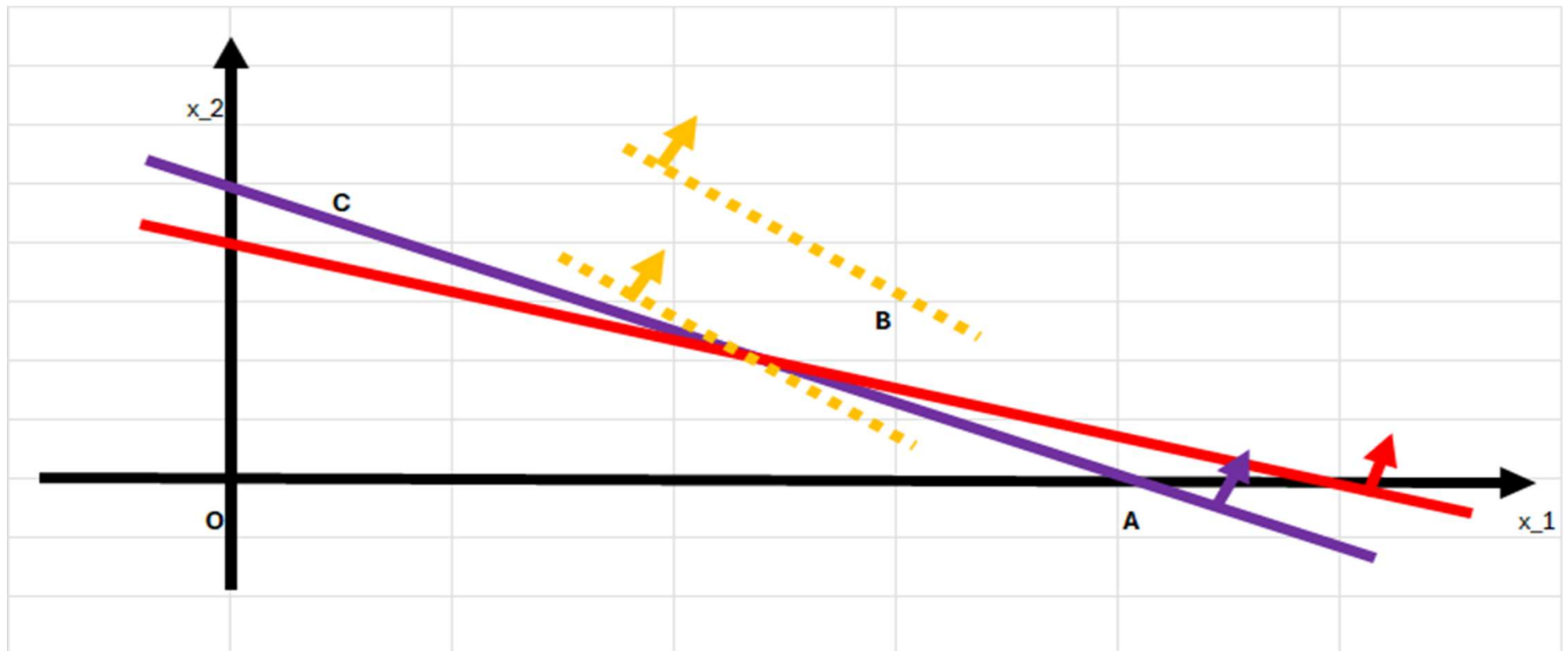
Operations Research

(Alternative/Multiple Solutions: Maximization Case)



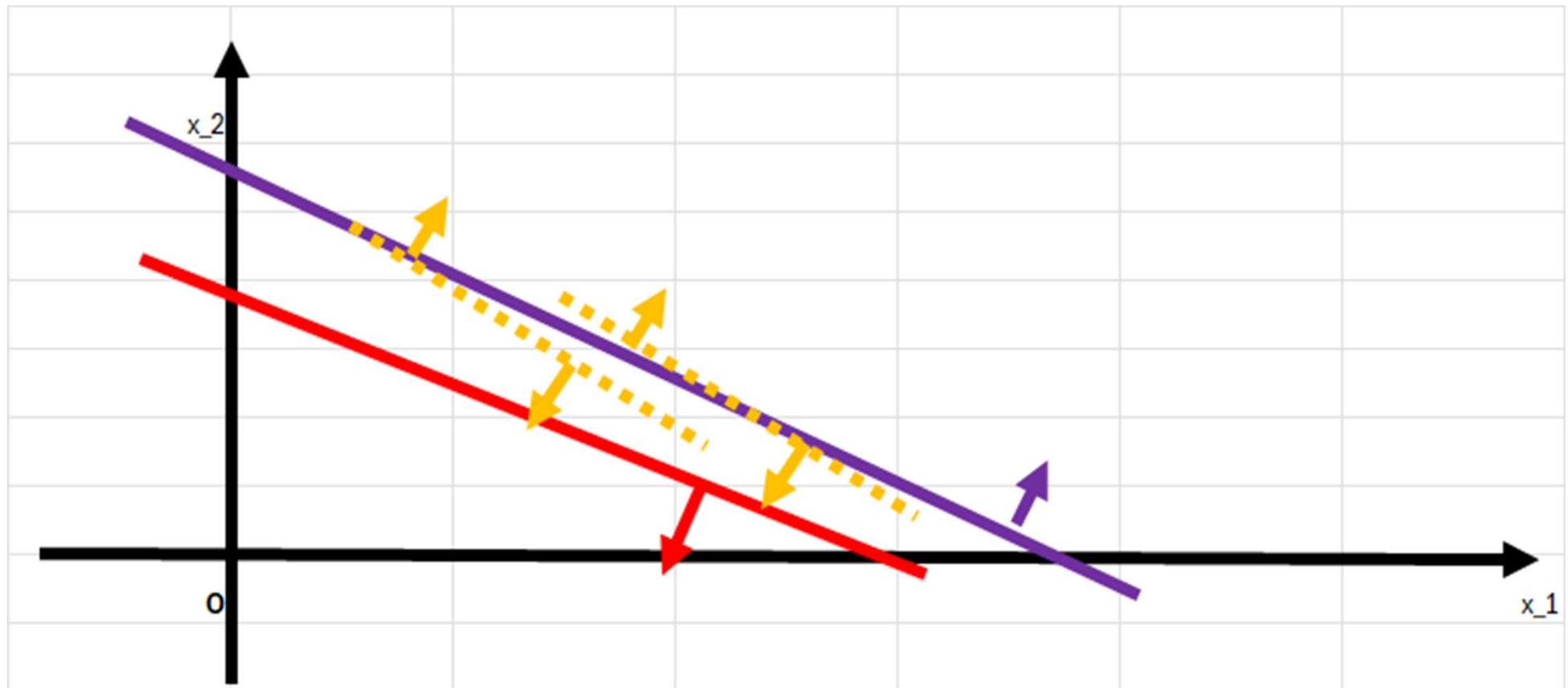
Operations Research

(Unbounded Solution: Maximization Case)



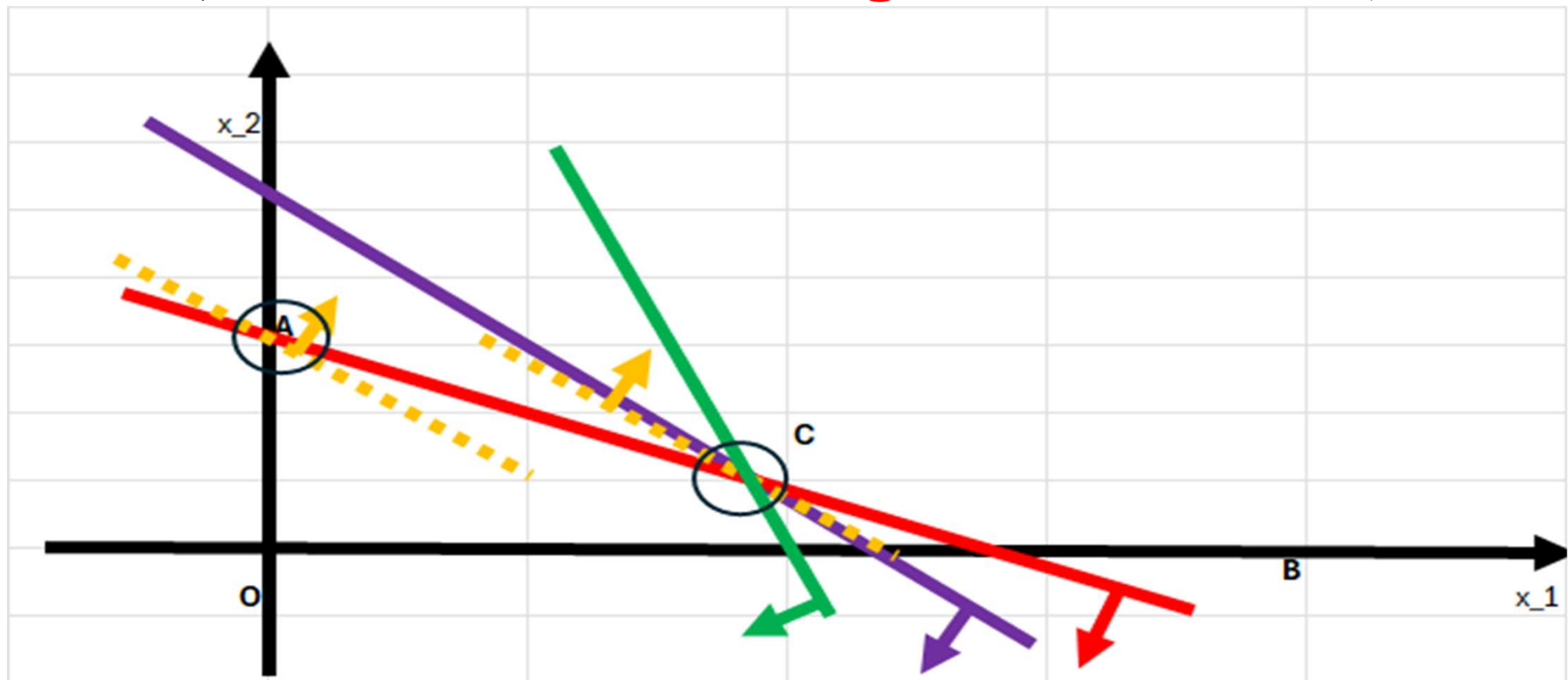
Operations Research

(Infeasible Solution: Maximization or Minimization Case)



Operations Research

(Infeasible Solution: Degenerate Solution)



Operations Research

(Example # 29 (using Tableau))

$$Ax = b \Rightarrow Ix = A^{-1}b$$

- Harpreet Singh & Associates produces both interior and exterior paints from two raw materials M_1 and M_2 .
- A marketing survey has revealed that the daily demand for interior paint cannot exceed that of exterior paint by more than 1 ton.
- Maximum daily demand of interior paint is 2 tons.

Operations Research

(Example # 29 (using Tableau))

Raw Material	Tons of raw material per ton of		Maximum daily availability (tons)
	Exterior Paint: x_1	Interior Paint: x_2	
M_1	6	4	24
M_2	1	2	6
Profit per ton ($\times 1000$)	5	4	

Operations Research

(Example # 29 (using Tableau))

$$\begin{aligned}\text{Max: } & +5x_1 + 4x_2 \\ \text{s.t.: } & +6x_1 + 4x_2 \leq 24 \\ & +1x_1 + 2x_2 \leq 6 \\ & -1x_1 + 1x_2 \leq 1 \\ & +0x_1 + 1x_2 \leq 2 \\ & x_1, x_2 \geq 0\end{aligned}$$

Operations Research

(Example # 29 (using Tableau))

$$\text{Max: } 1z - 5x_1 - 4x_2 - 0s_1 - 0s_2 - 0s_3 - 0s_4 = +0$$

$$\text{s.t.: } 0z + 6x_1 + 4x_2 + 1s_1 + 0s_2 + 0s_3 + 0s_4 = +24$$

$$0z + 1x_1 + 2x_2 + 0s_1 + 1s_2 + 0s_3 + 0s_4 = +06$$

$$0z - 1x_1 + 1x_2 + 0s_1 + 0s_2 + 1s_3 + 0s_4 = +01$$

$$0z + 0x_1 + 1x_2 + 0s_1 + 0s_2 + 0s_3 + 1s_4 = +02$$

$$x_1, x_2, s_1, s_2, s_3, s_4 \geq 0$$

Operations Research

(Example # 29 (using Tableau))

Basic	Z	X ₁	X ₂	S ₁	S ₂	S ₃	S ₄	Solution
Z	1	-5	-4	0	0	0	0	0
S ₁	0	6	4	1	0	0	0	24
S ₂	0	1	2	0	1	0	0	6
S ₃	0	-1	1	0	0	1	0	1
S ₄	0	0	1	0	0	0	1	2

Operations Research

(Example # 29 (using Tableau))

$$Ax = b \Rightarrow Ix = A^{-1}b$$

- $(1, -5, -4, 0, 0, 0, 0, 0) - (-5) \times \left(0, 1, \frac{2}{3}, \frac{1}{6}, 0, 0, 0, 4\right) = \left(1, 0, -\frac{2}{3}, \frac{5}{6}, 0, 0, 0, 20\right)$
- $(0, 6, 4, 1, 0, 0, 0, 24) \times \frac{1}{6} = \left(0, 1, \frac{2}{3}, \frac{1}{6}, 0, 0, 0, 4\right)$
- $(0, 1, 2, 0, 1, 0, 0, 6) - 1 \times \left(0, 1, \frac{2}{3}, \frac{1}{6}, 0, 0, 0, 4\right) = \left(0, 0, \frac{4}{3}, -\frac{1}{6}, 1, 0, 0, 2\right)$
- $(0, -1, 1, 0, 0, 1, 0, 1) - (-1) \times \left(0, 1, \frac{2}{3}, \frac{1}{6}, 0, 0, 0, 4\right) = \left(0, 0, \frac{5}{3}, \frac{1}{6}, 0, 1, 0, 5\right)$
- $(0, 0, 1, 0, 0, 0, 1, 2) - (0) \times \left(0, 1, \frac{2}{3}, \frac{1}{6}, 0, 0, 0, 4\right) = (0, 0, 1, 0, 0, 0, 1, 2)$

Operations Research

(Example # 29 (using Tableau))

Basic	Z	x_1	x_2	s_1	s_2	s_3	s_4	Solution
Z	1	0	$-2/3$	$5/6$	0	0	0	20
x_1	0	1	$2/3$	$1/6$	0	0	0	4
s_2	0	0	$4/3$	$-1/6$	1	0	0	2
s_3	0	0	$5/3$	$1/6$	0	1	0	5
s_4	0	0	1	0	0	0	1	2

Operations Research

(Example # 29 (using Tableau))

$$Ax = b \Rightarrow Ix = A^{-1}b$$

- $\left(1, 0, -\frac{2}{3}, \frac{5}{6}, 0, 0, 0, 20\right) - \left(-\frac{2}{3}\right) \times \left(0, 0, 1, -\frac{1}{8}, \frac{3}{4}, 0, 0, \frac{3}{2}\right) = \left(1, 0, 0, \frac{3}{4}, \frac{1}{2}, 0, 0, 21\right)$
- $\left(0, 1, \frac{2}{3}, \frac{1}{6}, 0, 0, 0, 4\right) - \left(\frac{2}{3}\right) \times \left(0, 0, 1, -\frac{1}{8}, \frac{3}{4}, 0, 0, \frac{3}{2}\right) = \left(0, 1, 0, \frac{1}{4}, -\frac{1}{2}, 0, 0, \right)$
- $\left(0, 0, \frac{4}{3}, -\frac{1}{6}, 1, 0, 0, 2\right) \times \frac{3}{4} = \left(0, 0, 1, -\frac{1}{8}, \frac{3}{4}, 0, 0, \frac{3}{2}\right)$
- $\left(0, 0, \frac{5}{3}, \frac{1}{6}, 0, 1, 0, 5\right) - \left(\frac{5}{3}\right) \times \left(0, 0, 1, -\frac{1}{8}, \frac{3}{4}, 0, 0, \frac{3}{2}\right) = \left(0, 0, 0, \frac{3}{8}, -\frac{5}{4}, 1, 0, \frac{5}{2}\right)$
- $\left(0, 0, 1, 0, 0, 0, 1, 2\right) - (1) \times \left(0, 0, 1, -\frac{1}{8}, \frac{3}{4}, 0, 0, \frac{3}{2}\right) = \left(0, 0, 0, \frac{1}{8}, -\frac{3}{4}, 0, 1, \frac{1}{2}\right)$

Operations Research

(Example # 29 (using Tableau))

Basic	Z	X ₁	X ₂	S ₁	S ₂	S ₃	S ₄	Solution
Z	1	0	0	3/4	1/2	0	0	21
X ₁	0	1	1	1/4	-1/2	0	0	3
X ₂	0	0	1	-1/8	3/4	0	0	3/2
S ₃	0	0	0	3/8	-5/4	1	0	5/2
S ₄	0	0	0	1/8	-3/4	0	1	1/2

Operations Research

(Example # 29 (using Tableau))

$$\blacksquare x_1^* = 3$$

$$\blacksquare x_2^* = \frac{3}{2}$$

$$\blacksquare s_1 = 0$$

$$\blacksquare s_2 = 0$$

$$\blacksquare s_3 = \frac{5}{2}$$

$$\blacksquare s_4 = \frac{1}{2}$$

$$\begin{aligned}\blacksquare f(x_1^*, x_2^*) &= 5x_1^* + 4x_2^* \\ &= 5 \times 3 + 4 \times \frac{3}{2} \\ &= 15 + 06 = 21\end{aligned}$$

Operations Research

(Example # 29 (using Tableau))

■ Constraint # 01

$$\begin{aligned} \diamond 6 \times 3 + 4 \times 3/2 + 1 \times 0 + 0 \times 0 + 0 \times 5/2 + 0 \times 1/2 &= 18 + 6 \\ &= +24 \end{aligned}$$

❖ Balances as raw materials M_1 **totally used**

■ Constraint # 02

$$\begin{aligned} \diamond 1 \times 3 + 2 \times 3/2 + 0 \times 0 + 1 \times 0 + 0 \times 5/2 + 0 \times 1/2 &= 03 + 03 \\ &= +06 \end{aligned}$$

❖ Balances as raw materials M_2 **totally used**

Operations Research

(Example # 29 (using Tableau))

- Constraint # 03

- ❖ $-1 \times 3 + 1 \times 3/2 + 0 \times 0 + 0 \times 0 + 1 \times 5/2 + 0 \times 1/2 = -3 + 3/2 + 5/2 = +1$

- ❖ Unbalanced ($x_2 \leq x_1 + 1$) as **at optimum level interior paint manufactured is 1.5 units less than that of exterior paint manufactured**

- Constraint # 04

- ❖ $0 \times 3 + 1 \times 3/2 + 0 \times 0 + 0 \times 0 + 0 \times 5/2 + 1 \times 1/2 = 3/2 + 1/2 = 2$

- ❖ Unbalanced ($+0x_1 + 1x_2 \leq 2$) as **at optimum level interior paint manufactured is 0.5 units less than that of permissible limit of 2.0 units**

Operations Research

(**PREPARATION** for **Simplex Method**)

- **Basic Feasible Solution (BFS)** chosen by transforming the linear program into **standard form**.
- Convert inequalities ($\leq$, $<$)/($\geq$, $>$) to equalities using **slack/surplus**.

Operations Research

(**PREPARATION** for **Simplex Method**)

- Select m variables (m is the number of constraints) to be **basic** (non-zero), and typically those are slack/surplus, which form identity matrix.
- Set all original decision variables to zero.
- Verify **feasibility** and ensure the resulting basic variables are non-negative.

Operations Research

(**STEPS** for Simplex Method)

- 1) Determine the starting **BFS**.
- 2) Select the **entering** variable using **optimality** condition. Stop if no entering variable (report result), else go to step # 03.
- 3) Select the **leaving** variable using **feasibility** condition.
- 4) Determine the new BFS by using **Gauss-Jordan** method. Go to step # 02.

Operations Research

(**OPTIMALITY** Condition for Simplex Method)

- 1) The **entering variable** in a **maximization** (**minimization**) problem is the **non-basic variable** having the most **negative** (**positive**) coefficient in z row.
- 2) The optimality is reached at the iteration where **all** the z row coefficients of the non-basic variables are **non-negative** (**non-positive**).

Operations Research

(**FEASIBILITY** Condition for Simplex Method)

- 1) For both **maximization** and **minimization** problem the **leaving** variable is the basic variable associated with the **smallest non-negative** ratio (with **strictly positive denominator**).
- 2) For ties for ratios are broken arbitrarily.

Operations Research (Lagrangian Multipliers)

- **Lagrange** multipliers can be used to solve **Non-Linear Programming** (NLP) problems in which all the constraints are **equality** constraints.
- Example of NLP

$$\begin{aligned} &\text{Max/Min: } f(x_1, \dots, x_n) \\ &\text{s.t. } g_j(x_1, \dots, x_n) = b_j, j=1, \dots, m \\ &\quad x_1, \dots, x_n \geq 0 \end{aligned}$$

Operations Research (Lagrangian Multipliers)

- Assign **Lagrangian Multipliers**, $\lambda_j \geq 0, j=1, \dots, m$, with each constraint which gives us the Lagrange equation of the form $L(x_1, \dots, x_n, \lambda_1, \dots, \lambda_m) = f(x_1, \dots, x_n) + \sum_{j=1, \dots, m} \lambda_j \times [b_j - g_j(x_1, \dots, x_n)]$
- Remember Lagrangian method turns the problem into an **unconstrained** problem, however, the optimal solution $(x_1^*, \dots, x_n^*)$ must still satisfy the original constraints.

Operations Research

(Lagrangian Multipliers)

- In case of **maximization** (**minimization**) first ensure that $\partial L / \partial x_i = 0 \quad \forall i=1, \dots, n$ and $\partial L / \partial \lambda_j = 0 \quad \forall j=1, \dots, m$
- $\partial L / \partial x_i = 0$, implies $\partial f(x_1, \dots, x_n) / \partial x_i - \lambda_j \partial g_j(x_1, \dots, x_n) / \partial x_i = 0, \quad \forall i=1, \dots, n$
- $\partial L / \partial \lambda_j = 0$, implies $b_j - g_j(x_1, \dots, x_n) = 0, \quad \forall j=1, \dots, m$
- Check the Hessian matrix and the condition, thereof for maximization (minimization)
- Proper substitutions will yield the optimal solution

Operations Research

(Lagrangian Multipliers: Example # 30)

$$\begin{aligned}\text{Max: } f(x_1, x_2) &= -2x_1^2 - x_2^2 + x_1x_2 + 8x_1 + 3x_2 \\ \text{s.t. } 3x_1 + x_2 &= 10 \\ x_1 &\geq 0, x_2 \geq 0\end{aligned}$$

- $L(x_1, x_2, \lambda_1) = -2x_1^2 - x_2^2 + x_1x_2 + 8x_1 + 3x_2 + \lambda_1 \times [10 - 3x_1 - x_2]$
- $\partial L / \partial x_1 = -4x_1 + x_2 + 8 - 3\lambda_1 = 0$
- $\partial L / \partial x_2 = -2x_2 + x_1 + 3 - \lambda_1 = 0$
- $\partial L / \partial \lambda_1 = 10 - 3x_1 - x_2 = 0$ (reduces to the constraint)

Operations Research

(Lagrangian Multipliers: Example # 30)

- Solving the first and second equation yields $x_1 = 19/7 - \lambda_1$ and $x_2 = 20/7 - \lambda_1$
- Substituting these two values in the third equation ensures that we get: $\lambda_1 = 1/4$; $x_1 = 69/28$; $x_2 = 73/28$
- $H(x_1, x_2) = \begin{bmatrix} -4 & +1 \\ +1 & -2 \end{bmatrix}$ and as $\det(H) = 7 > 0$, hence it is concave which proves the idea of maximization
- $\lambda_1 = 1/4$ should be remembered as the rate of increase of the objective function so long as the constraint is not violated

Operations Research

(Karush Kuhn Tucker (K-K-T Conditon))

- K-K-T condition gives the **necessary** and **sufficient** conditions for the optimal solution in case of NLP

$$\begin{aligned} &\text{Max/Min: } f(x_1, \dots, x_n) \\ &\text{s.t. } g_j(x_1, \dots, x_n) \leq b_j, j=1, \dots, m \\ &\quad x_1, \dots, x_n \geq 0 \end{aligned}$$